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Face Matching in Indian Citizens using CNN

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Abstract—**Image processing is a way of performing certain image operations in order to get the desired output from the image. There are numerous applications that use the processing of images. There are several techniques in image processing, algorithms that can be used to achieve expected performance for different applications. One of the most trending studies is facial recognition and classification using image processing. Because everything in this era is digitalized, face recognition is a vital part of it. While the subject is well studied, there are still many difficulties faced, such as lighting changes, different noises or other variables. In this paper we are concentrating on face matching of Indian citizens i.e. to accurately recognize if it is the same person having considered the pictures of them at different ages. We have collected data sets of Indian citizens, pictures of each person from young age to recent ones. Vggnet is used for feature extraction and prediction. Objective for the proposed method is to get good accuracy rate and also to match the pictures correctly providing proper matching rate. The motivation for this project is to help in legal system especially in crime branch to track down guilty and also to group a huge data into that particular person's file.**

Keywords— ***Face matching, feature extraction, deep learning, system accuracy, VGGNet, CNN***.

I. Introduction

In the modern age, facial recognition, detection and matching play an essential role. It's a part of the vision of computers. It has different smart surveillance applications and also helps to track offenders and numerous other applications for law and order. Three primary stages of image processing, feature extraction, and classification and identification are involved in face recognition and matching. Deep learning in the area of face detection and recognition has shown great results in recent years. The conventional way of processing images involves scaling, cropping and modifying the input image and then further processing it for extracting features, but this is automated with the use of Vggnet, cropping, aligning and adjusting images to a standard size and identifying and extracting features all are done automatically using this method.

Deep learning approaches have recently sparked interest due to their ability to surpass prior state-of-the-art algorithms in a number of tasks, as well as a large amount of complicated data from a variety of sources. (e.g., Sensory, auditory, medicinal, social, and visual) [1]. There have been a few previous works on facial detection and recognition of attributes, such as a brand-new deep system that predicts facial characteristics and it's used as a gentle modality to enhance face recognition accuracy. [3] To recognise face expressions, a VGGnet deep cnn is used. With a more extensive network design and The recognition rate is significantly increased using a 3 * 3 small convolution kernel and a 2 * 2 small pool kernel, and the number of parameters is just minimally more than the shallow layer. [9] To achieve a high recognition system rate with minimal processing time and resources, computer vision requires powerful categorization methods. In this paper, we suggest a system that matches the face of the individual and agrees that the feedback given is of the same person, here age based photographs are considered, the system is trained in such a way that even if photograph of a person aged at the age of five is given or the same person's photograph at the age of 25 is given, the system provides an accuracy saying that both are same person and displays their name with accuracy rate of match. With a minimum of five photographs of each person, we have collected a dataset of 50 Indian citizens that gives total of 250 photographs in which 200 images are utilised for training and 50 images are used for testing.

In such projects, facial features play a significant role, factors are required to identify a face or to distinguish them. Eyes, nose, ear length, width, and the spacing between these characteristics are attributes. They are constant and they don't change. For feature extraction, identification, and prediction Vggnet is used here. The objective is to provide a framework that correctly identifies the individual where input may be from any age group of that person. The system can be very helpful in tracking down criminal if we have that person's any age group photograph, the system can match with the suspect and give the accuracy of the match and the proposed system can also be helpful when there is huge set of images or photographs to be sorted and to do it manually it will consume a lot of time so using this system it can be done automatically. The motive of this paper is to help the legal system specially to track down the criminals i.e. for example if the person is a criminal and there is a picture of him in the database which is 20-year back saved one, and the same criminal's picture is taken now and he is of 60 years, using the proposed method we can obtain the matching rate of the two pictures and find out the guilty. This research concentrates on developing a system that mainly concentrates on getting the matching rate of the person's picture at different age, i.e. to check if the given input picture of a 60-year-old for example, provides matching rate and predicts it's the same person from the database fed to it but with pictures of the person at the age of 5 or 10 or 20. This can help in tracking a person and even classifying each person into different folders of their own based on matching thus providing novelty to the paper.

II. Literature Review

Athanasios Voulodimos et al. [1] have conducted a deep learning study because it is common in science and provides promising results in computer vision. Here they have addressed various techniques including CNN, Deep Boltzmann Machines and Belief Networks, and Autoencoders Stacked Denoising Autoencoders. The paper is divided into four sections that deal with object identification, facial recognition, behavior and activity recognition, and human pose estimation, measured all the odds and checked which approach best fits the applications given. The authors conclude by noting that the challenges in methodology will continue to attract the research community in the coming years. Qiong Cao et al [2] proposed a paper on the dataset used in face recognition across various poses and ages. They have gathered publicly accessible datasets and implemented a wide scale dataset named VGGFace2, their aim was to use the maximum number of images with different poses and variations. Paper describes how the dataset, in particular the automatic and automated data set, was obtained and manual steps of filtering to ensure that the photographs of each identity are highly accurate. Method used is CNN, they have trained dataset using ResNet-50 CNN and have concluded that using this method on VGGFace2 achieves a good result. Fariborz Taherkhani et al [3] suggested an end-to-end deep network with improved performance for predicting facial traits and identifying face photos simultaneously. Modality blends predicted identity attributes with face modality features to increase face recognition performance in their model, which employs a shared CNN feature space to train these two tasks together. Methods provided by them is compared to a number of competing algorithms, including PANDA, Face Tracer, MT-RBM-PCA and ANet+LNet for attribute prediction. FaceTracer extracts handicraft features from a functional face image region, such as color histogram and HOG, and concatenates them to train an SVM classifier that predicts attributes. When compared to when the model is trained individually, the approach increases face recognition as well as face attribute prediction.

Guodong Guo et al [4] have taken into account both deep learning architectures and unique recognition problems such as lighting variation, pose variation, low resolution, etc., they provide a full overview of face recognition approaches using deep learning. They have primarily found the CNN approach and have obtained excellent results for still photographs, videos and heterogeneous face matching for cross model FR. In this survey they have done step by step process for face recognition and concluded that though recognition accuracies have been improved there are still challenges faced in the practice. Vi-Wen Chen et al [5] have proposed a classification of age groups based on images, a system for estimating three age groups: infant, adult and senior citizens. Following the detection of a face in the image frame obtained, the Image alignment algorithm Lucas-Kanade is used to extract the human facial region and locate 52 feature points. These feature points, as well as the facial region they correspond to, are used to generate a model of an active look (AAM). The texture features are fed into a support vector machine (SVM) after facial image warping to assess the level of age grouping. Ashwini Jalindar Mawale et al [6] have proposed a system that uses Gabor filter to identify the wrinkles in the face. Their main goal is to increase detection rate while lowering miss ratio. Using the Gabor filter, they were able to detect facial wrinkles. They have used hessian filter to detect the curve lines. The proposed algorithm produces visually better results than current system results, according to experiments. Cons include the fact that they did not recognize the entire face photograph as input and instead selected a section of the face as input image, such as the forehead, under the eye, cheek portion, and so on. Klemen Grm et al [7] Using the Labelled Faces in the Wild dataset, researchers investigated the impact of numerous factors on the performance verification of four contemporary CNN models. Based on their comprehensive testing, they investigated the merits and disadvantages of deep learning models. The versions being evaluated include AlexNet, VGG-Face, GoogLeNet, and Squeezenet. The consistency of deep face recognition models is determined by a number of variables (or covariates) that can be classified into various groups. Model-related covarieties and image quality are the covarieties considered here. When compared to lower-resolution photos, than those with which they were taught, they all decline quickly and substantially. With the right design and training techniques, a deep learning model can be made fairly resilient to common sources of image-quality degradations. Amit Dhomne et al [8] they improved the previously used methodology and obtained a lot of precise results by using D-CNN, VGGNet architecture for gender recognition. Benifit of using DCNN is that it automatically extracts the function from an image and outputs it, as well as increasing geometric transformations and deformations, as well as 2-dimensional shape and figure changes. No one had tried VGGNet for prediction of gender with the Celebrity face dataset before, and no one had gotten such good results. The code for the Graphical Processing Unit was written in Python using OpenCV, Pytorch, and Tensorflow libraries. He Jun et al [9] present a recognition of face expressions algorithm based on the VGGNet deep convolutional neural network, with the goal of improving on traditional convolutional neural networks' low recognition rate in the facial expression database. The recognition rate is greatly enhanced with a 3 * 3 small convolution kernel and a 2 * 2 small pool kernel with a deeper network design. The device performs well in the FER2013 database when it comes to recognition. They employed a The convolutional layer is separated into five segments, each with 2-4 convolutional layers, and each segment is connected to a max-pooling layer at the end to reduce image size in this paper's 19-layer VGGNet network model. The accuracy of the system is 73.06 percent. Ze Lu et al [10] have presented a paper for improving low-resolution image facial recognition. They also primarily concentrated on photographs taken from low-resolution CCTV video. Face photographs of three different resolutions were used to train a trunk network and two branch networks for the system. The trunk network extracts discriminative features that are resistant to resolution shifts. As resolution-specific paired mappings, two branch networks, both HR-trained and LR-targeted, operate, transforming HR and LR functions into a zone in which their differences are reduced. The proposed methodology is tested on the LFW and SC face datasets, demonstrating that it is both reliable and superior to current methods.

Mohannad A. Abuznied et al [11] employing In this study, they present a new technique for improving human face recognition using a Back-Propagation Neural Network (BPNN) and feature extraction based on the training pictures' correlation. A significant addition of this study is the construction of the The T-Dataset is a subset of the original training dataset that was used to train the BPNN. The key benifaction of this research is the use of LBPH, multi-KNN, and BPNN to improve human face recognition. The strength of their technique stems from the addition of a step later feature dimension reduction and extraction to produce a unique The T-Dataset will be used to train the BPNN. The accuracy obtained by their method is 98%. Ketan Kotwal et al [12] to recognize facial makeups, a deep learning-based presentation attack detection (PAD) approach is presented by the authors. They propose using a convolutional neural network (CNN) to extract characteristics that can tell the difference between presenting and presentations with and without age-related facial makeup. They provide AIM (Age Induced Makeups), a new dataset containing There are more than 200 video Demonstrations of authentic old-age makeups. The median matching results of a recent CNN-based FR system are reduced by 14% as a result of AIM makeups in their tests. They demonstrate that a CNN that has been trained for FR tasks is an effective feature extractor for recognising makeups without the need for explicit transfer learning or fine-tuning. The accuracy rate is 98% from their method. Hongyu Pan et al [13] in this research, they propose a new loss function for a more accurate age estimation using mean-variance loss, it is a term used to describe a type of distribution learning. The mean-variance loss is made up of two parts: a mean loss that penalises the difference between the estimated age distribution's mean and the ground-truth age, and a variance loss that penalises the estimated age distribution's variance to ensure that the dispersion is focused with precision. For age estimation, Convolutional Neural Networks incorporate the suggested softmax loss with mean variance loss (CNNs). During age estimate, the proposed loss is effective for producing a yet concentrated accurate label distribution estimation. Experiments on the MORPH II, CLAP2016, FG-NET and AADB databases reveal that their strategy outperforms the competition and Does a good job of generalising to image aesthetics evaluation tasks. 83.72% is the accuracy got from their method. Shifeng wu et al [14] work investigates the elements that influence face recognition, focusing on the impact on identification verification based on age and gender outcomes, as well as the use of the To classify facial traits, a deep learning algorithm is used. The method used is MTCNN. The impact of age and gender on classification outcomes is investigated in this research. The findings reveal that middle-aged males have a lesser recognition effect in male participants than youth and the elderly. They show that the recognition rate for young males is 84 percent., The male middle-aged group has a recognition rate of 71%, and The male senior age group has an 83 percent recognition rate, demonstrating that The male youth and senior age groups have a stronger recognition effect, but the recognition effect of the middle age group is poor. The female youth group has a recognition rate of 71 percent, 73 percent for the females in their thirties, and 77 percent for the elderly women's group, demonstrating that as people get older, the impact of recognition grows. Women's recognition effects very little by their age. Males are more likely to be recognised than females. The precision is 83.73 percent.

Soon-Gyo Jung et al [15] use several datasets to assess the accuracy of four commonly used facial recognition technologies, including Face++, IBM BV Recognition, AWS Recognition, and Microsoft Azure Face API, in inferring user features such as gender, race, and age. Despite the fact that the study used a high-quality dataset, only one of the four tools can be used, i.e. Face++, can perform this task with a rate of accuracy of greater than 90%. Age inference looks to be a difficult task, as all four techniques scored poorly. They combine photos of well-known celebrities with real-world demographic data. In addition to the images, they have gathered the celebrity's birthdate and the date the snapshot was taken. Because of the generally good image quality and the presence of ground truth in terms of age, gender, and race, celebrities' photos were chosen. Ass far as the accuracy goes, it shows that Face++ gives 90% accuracy compared to other tools. Felix Anda et al [16] present the results of a review of current Services for predicting cognitive age. The goal of the The purpose of the evaluative and comparative examination of the various services was to uncover trends and issues with their performance. The noticeable shortage of suitable sample photos in specified age groups is one key a barrier obstructing the accurate development of the services under consideration. To address this problem, a dataset generator was created that combines numerous unbalanced datasets to create a well-balanced, chosen group of digital pictures with age and gender annotations. They have considered Kairos, AWS, DEX and Azure for evaluation. They concluded that the job of choosing the Face age estimation with the finest results methodology is more complicated than simply picking the one which has the lowest overall mistake rate, because the efficacy of all four treatments investigated is influenced by gender and age.

TABLE I. SUMMARY OF LITERATURE SURVEY WITH THEIR MODELS

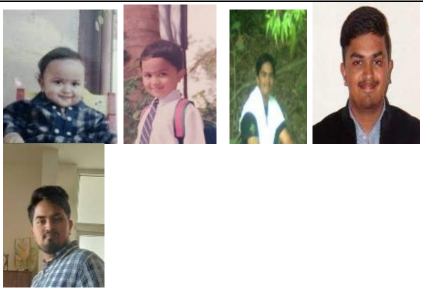
Sources	***Features selected***	***Models selected***	**Accuracy (in %)**
Qiong Cao(et al).	1.Pose variation 2.Age estimation	RESNet VGGface SENet	87
Ze Lu(et al)	Low-resolution images	SVM Deepcoupled RESNet	78.89
He Jun(et al)	Facial expression	CNN VGGnet	73.06
Fariborz Taherkhani(et al)	Facial attributes	CNN VGG 19	78.82
Amit Dhomne(et al)	Gender prediction	VGGNet D-CNN Soft-max MLP	95
Ashwini Jalinder Mawale(et al)	1.Facial 2.Wrinkles 3.Curvelines 4.Morphology	Gabor Filter Hessian Filter	83.43
Mohannad A. Abuznied(et al)	Facial expression Lighting variation	KNN BPNN LBPH	98
Ketan Kotwal (et al)	Age related facial makeup	PAD CNN	98
Hongyu Pan (et al)	Age estimation	Mean variance loss CNN	83.72
Shifeng wu (et al)	Face recognition Impact of age and gender.	MTCNN	83.73

III. PROPOSED SYSTEM

A. Dataset description

Dataset collected for the project is that of images of Indian citizens, the dataset collected is that of the average age group of 19 and average of 4 images are collected from every individual that show them in different age group or at least having difference of 5 years, this is done to ensure recognition accuracy for that person's image. A sample of 100 people's images are collected that leaves with 400 images in total. The individuals were asked to share their pictures from the age of 5 till their current age with each picture having difference of at least 5 years in them, we could collect images of individuals who are in the age group of 20-30 mostly as they possess several images from their young age till now, the images acquired vary from person to person in number, we received average of 4 images from group of 35 individuals, average of 5 from group of 30 individuals, average of 6 from group of remaining 35 individuals and also not everyone could give the images with exactly 5 years difference some of them gave images with difference of 6-7 years or 3-4 years. There are 300 photos used for training and 100 images used for testing. Though there are certain features that do not change even if the person grows older like distance between ears and nose, distance between eyes etc., but might take a slight variation in decimal values, so when there are pictures of individual at different age are available it becomes convenient to calculate the whole average and find the nearest matching value of the input image to that of the average obtained. Like this we can get the accuracy rate increased for the image if we have more number of training dataset available of a single person. Given below is the table that depicts the sample of individual pictures, from their young age till the recent one with at least a gap of 5 years between the pictures. In the first column, there is a sample picture of the individual and the second column consists of the five images of the same individual from different age groups showing their growth differences, images have pose variations and light variations in each individual's sample

TABLE II. SHOWING THE DATASET COLLECTED, FIRST COLUMN CONTAINS SAMPLE OF EACH INDIVIDUAL AND SECOND COLUMN SHOWS DATASET OF EACH INDIVIDUAL AT DIFFERENT AGES

Sample	***Images At Different Age***
Person1 	
Person 2 	
Person 3 	

B. CNN

Convolution neural network is used in this paper as it is a multi-layered neural network used primarily for image recognition, extraction of characteristics, segmentation, and other associated data as well. Biological processes influenced

Convolutionary networks The communication pattern between neurons is similar to the organisation of the animal's visual cortex. Each neuron takes different amounts of inputs and passes through with activation function and responds with an output. In the proposed system CNN is used to extract and predict facial features, with features as inputs which act as weights and are fed into feed forward network which then predict the output. A convolutional neural network has three types of layers: convolutional, pooling, and fully connected. Each of these levels has its own set of settings that may be tweaked, and each does something different with the input data. Among these, The convolutional layer is mostly used to extract features; The convolution kernels are in charge of extracting various features. The activation function converts the convolutional layer's output as a nonlinear function The pooling layer's job is to lower the parameters and compress the image without affecting image quality. The fully-connected layer's job is to turn the feature graph of the convolution layer into a fixed-length feature vector. Local connection and weight sharing are two properties of convolutional neural networks. Because Local connection refers to the non-complete connection of neurons between nearby layers, where each node in the convolution layer only receives a little amount of the top layer's input. Weight sharing means that all parameters within each convolution kernel are shared over the whole graph, and the convolution kernel's weight coefficient does not change as a result of different picture positions. CNN's main idea is to use small filters to investigate spatial features. As we progress deeper into the network, the spatial dimension will steadily decrease. The depth of the feature maps, on the other hand, will increase.

C. VGGNet

VGG is applied in this process, it is a specific object recognition framework that supports up to 19 layers and is the most commonly used image recognition architecture. There are three layers in it: an input layer, a concealed layer, and an output layer.. A 224x224 pixel RGB image is taken by VGG and transformed into a standard size to keep the size of the input image constant. The use of Using pre-trained data to set parameters in a certain layer enhances the network's accuracy and speeds up convergence. A stack of convolution layers is used to send the training images through. In the VGG16 architecture, There are 13 convolutional layers in all, with three completely connected layers. Instead of having huge filters, VGG features smaller filters (3*3) with better depth. The first two layers are 3*3 convolutional layers, and the first two layers include 64 filters, resulting in a volume of 224*224*64. 3*3 filters with a stride of 1 are always used. Following that, a pooling layer with a maximum pool size of 2*2 and a stride of 2 was employed to lower the volume's height and breadth from 224*224*64 to 112*112*64. After that, there are two further convolution layers consisting of 128 filters. The newly obtained dimension is 112*112*128 as a result of this. The volume is reduced to 56*56*128 after using the pooling layer. Two further 256-filter convolution layers are addedAfter that, a The size is reduced to 28*28*256 by using a down sampling layer. A max-pool layer separates two additional stacks, each having three convolution layers. A Fully Connected (FC) layer with channels 4096 in number and a softmax output of classes of 1000 is created from the 7*7*512 volume after the final pooling layer. Fully-connected layers are converted to convolution layers during the network test phase, allowing any high-resolution test photographs to be obtained for later model testing.

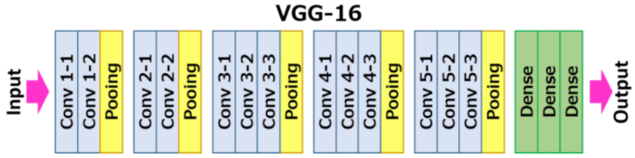


Fig 1 VGG16 Model Architecture

D. Complexity of the proposed System

The proposed method requires a dataset of at least 4 pictures from each individual to train the system, this was a tedious job to get the individuals who had pictures from their young age till their recent years with at least a gap of 5 years between each picture. We overcame this difficulty by only collecting the data from citizens below the age of 35 as they would have the record of their pictures from every stage of their age. We even got the pictures from senior citizens and that of middle aged citizens showing up with 3-4 pictures from their young age which became useful to test the system's reliability of whether it can match the picture of the person at their old age with that of the young age picture present in the database.

E. Model Design

The system is trained with the same person's images of different age groups so that when a test image is presented, it can predict the performance. The method used here is Vggnet for identification of facial features, prediction of the output. The Vggnet mainly consists of three layers the convolution layer which has 13 layers of convolution, pooling layer which is 2x2 maxpool and there is fully connected layer. The architecture consists of two layers input layer where the image is given and then second layer consists of hidden layer where feature extraction happens and threshold and weights are obtained from all the images and are trained and when the test image is given as input then the system compares it with the trained sets and gives the output and predicts the accuracy of match. Finally output image is given with matching rate and also the top 3 images from which the most features were obtained.

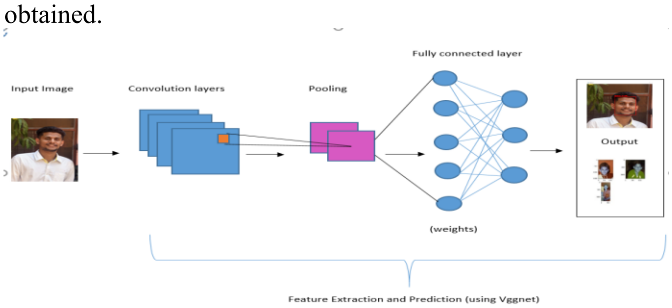


Fig. 2. Architecture Diagram

F. Comparison with other Techniques.

Considering the paper that He Jun [et al] [9] proposed which detects the facial expression using Vggnet, they have considered Vggnet network model with 19 layers, where the Each segment of the convolutional layer contains 2-4 To reduce image size, each segment is paired with a max-pooling layer at the conclusion of the convolutional layers. In this research, VGG19+dropout+10crop+softmax is used to train and evaluate the FER2013 database. The accuracy achieved by them is 73.06%. In the other paper by Ze Lu [et al] [10] have proposed RESNet with deep coupling for low-resolution face recognition. Two branch networks and one trunk network make up the architecture. The trunk network is intended to extract distinguishing facial characteristics that are resistant to face picture resolution degradation. Two branch networks are used to learn coupled mappings that reduce the distance between feature vectors extracted from an HR image by the trunk branch and feature vectors extracted from an LR image by the trunk branch that corresponds to it. The system accuracy obtained by them is 78.89%. Our system proposes a method to predicts the matching rate of the picture of the individual at any age provided it has trained images of the same person's images from early age or vice-versa. Proposed system uses Vggnet in cnn, Vgg16 has three layers namely convolution, pooling and fully connected layers.It has 13 convolution layers and 2 fully connected layers. The accuracy of the proposed system is 83.6%.

TABLE III TABLE SHOWING THE COMPARISON WITH OTHER EXISTING TECHNIQUES.

Sl.No	Objective	Method used	Accuracy
1	Facial expression detection	VGG-19	73.06%
2	Low resolution face recognition	RESnet	78.89%
3	Face Matching	VGG-16	83.6%

IV. RESULT

The proposed methodology takes an image as input and then output is given which shows the accuracy rate of the face matching. As shown in the architecture testing phase takes the given images and process them using the Vggnet which help in identifying the face and also detect the facial features.

Fig 3 Sample to show how the VGGNet work in detecting the face and the features to be extracted.

The figure 3 depicts pictures of the same individual taken at various ages., as both the methods are prominently used for face detection and also facial features extraction, a sample is put up to show few examples of features detected, here in this basic features like eyes, nose, mouth are detected as well as the distance between the ears, distance between the ears and nose, the distance between eyes etc. are also calculated and those measurements act as weights in the Convolution neural network. Then these extracted features are fed in CNN as weights to calculate the average outcome from these training images. Then in testing phase the input image is given which needs to be tested and CNN then calculates the feature measurements, checks with that of got by the training images and matches the image which closely or perfectly matches the weights of the training set and provides accuracy rate of matching.

TABLE IV TABLE WITH INPUT IMAGE, OUTPUT WITH ACCURACY AND IMAGE OF TOP 3 IMAGES FROM WHICH FEATURES ARE MOST EXTRACTED.

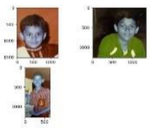
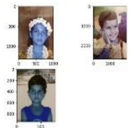
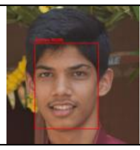
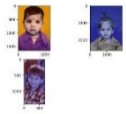
Input image	Output with prediction rate	Top 3 images from which features are most extracted
		
		
		

TABLE V. RESULT OF TESTING IMAGES

Batch	Training Images	Testing Images	Accuracy (f-measure in %)	Recall	Precision
1	400	200	83.6	0.76	0.94
2	175	125	87.5	0.77	0.82
3	205	183	85	0.68	0.84

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The accuracy of the system is obtained by using confusion matrix. An NxN matrix is used to evaluate the N is the number of target classes, and performance of a classification model. The matrix compares the actual goal values to the predictions of the machine learning model. This provides us with a comprehensive picture of how accurate our classification system is working and the types of errors it makes. Here in the above table the images are divided into batches consisting of varying numbers. The first batch shows 500 images for training the system and 200 for testing, similarly other two batches are done with varying number of testing and training images. This is done to make certain that whether the system gives good accuracy when it is trained with varying number of images.

V. Conclusion

There are systems which predict the person's age or even match the face but there are very few papers or systems that are dedicated to matching the face or finding the similarity rate keeping the images of person at different ages and then give an input of the same person at any random age and get the results of accuracy rate. This paper proposes a method for identifying people by comparing their images to previously trained images of the same individual at various ages, for example if an image of 20-year-old person is given as an input then it shows hoe much that person matches the similarity provided training images of the same person at different ages like 5, 10, 15 years etc. The system uses Vggnet to detect and identify the features and predict the outcome. The dataset collected is that of Indian citizens and average of 4 photos is collected from each individual presenting them at different ages. The system first trains the given images and features are extracted and passed on as weights to cnn which then calculates the average which then is compared with that of the testing image measurements and then the closest value or matching value is put forth as the similarity rate and the output is given which is matching rate of that person's image along with the top three images which have contributed the most in feature selection. While existing systems consider face recognition, age classification, gender classification as different projects or might combine two or three aspects together and provide a system, there are quite few papers on face matching even if there are they concentrate on matching the faces with faces already stored faces in database and do not consider the images of the same person from different ages, our system concentrates on providing a system that can match the person's old picture with the recent one predicting that it is the same person with the matching rate. The objective is to provide a better and accurate face matching system concerned with age, to get a good result for matching rate of the person's picture at different ages so that it would be helpful in many fields. Further work can be done by collecting more number of dataset and show what are the variations that happen in the facial structure and features while aging and plot a graph for it.

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